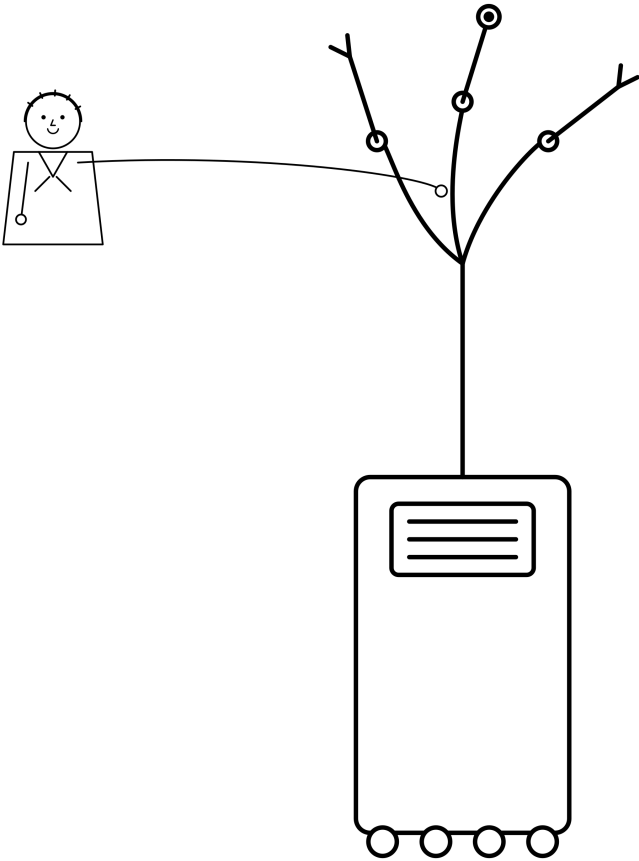


Patient-Robot Instructions: AI Oncology Trials - Surgical Robots

Patient-Robot Interaction: Surgical Robots

Adult Oncology Trial Setting



Surgical Robots



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Surgical Robots

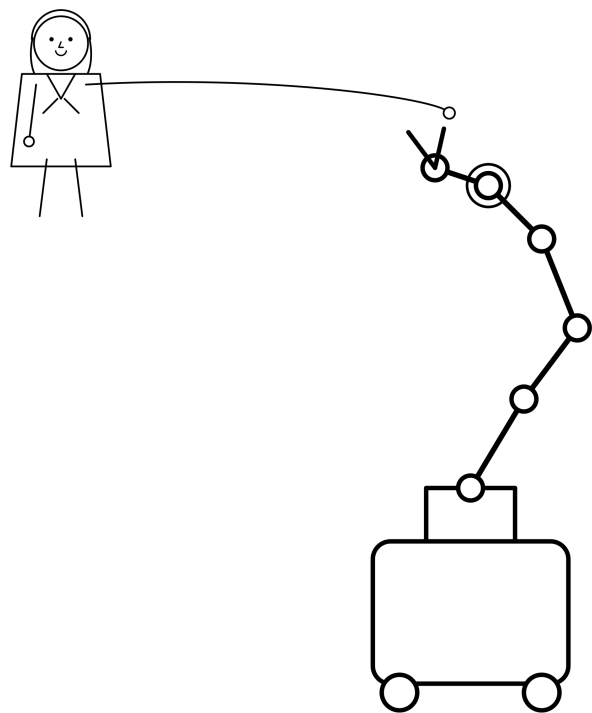
This surgical robot assists your surgeon in performing a minimally invasive robotic prostatectomy through small incisions using precise multi-arm instruments.

1. Lie face-up on the operating table, place both arms flat on the padded arm rests with palms up, and remain still while anesthesia is administered over approximately 2 minutes.
2. You will be under general anesthesia for the entire 90–180 minute procedure; the robot's 4 arms operate through 8–12 mm incisions while the surgeon controls it from a nearby console.
3. After the robot arms retract, you are moved to recovery where you will awaken in 15–30 minutes; remain lying down until a nurse confirms your vitals.

Patient-Robot Instructions: AI Oncology Trials - Cobots

Patient-Robot Interaction: Cobots

Adult Oncology Trial Setting



Cobots



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Cobots (Collaborative Robots)

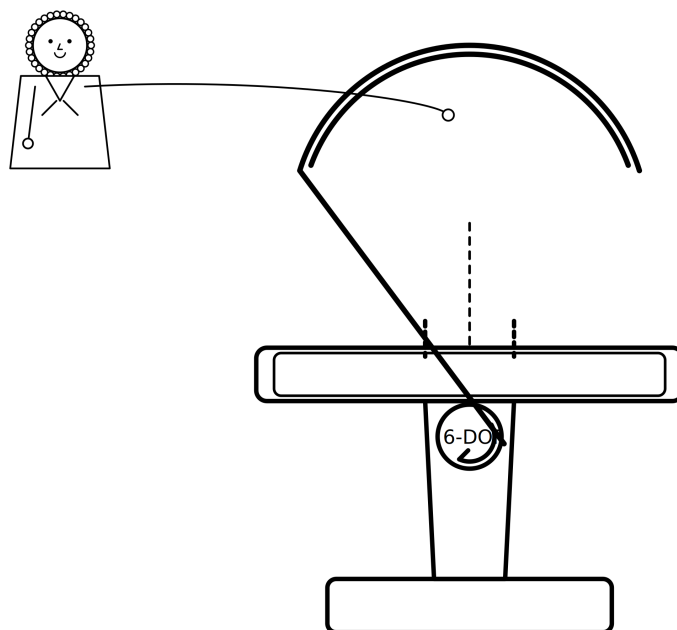
This collaborative robot arm gently positions your arm and assists with automated tissue sample collection for your breast cancer biopsy analysis.

1. Sit in the adjustable exam chair, rest your arm on the padded support with your palm facing down, and keep your hand relaxed and fingers open for approximately 1 minute.
2. The cobot moves at a maximum of 250 mm/s with force-limited contact; hold still for 10–25 minutes while it repositions your arm 2–3 times and collects samples, saying “Stop” if you feel discomfort.
3. The cobot retracts to its home position; remain seated for 2 minutes while results are reviewed, then receive a bandage at the collection site before leaving.

Patient-Robot Instructions: AI Oncology Trials - RT Positioning Robots

Patient-Robot Interaction: Radiotherapy Patient-Positioning Robots

Adult Oncology Trial Setting



Radiotherapy Patient-Positioning Robots



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Radiotherapy Patient-Positioning Robots

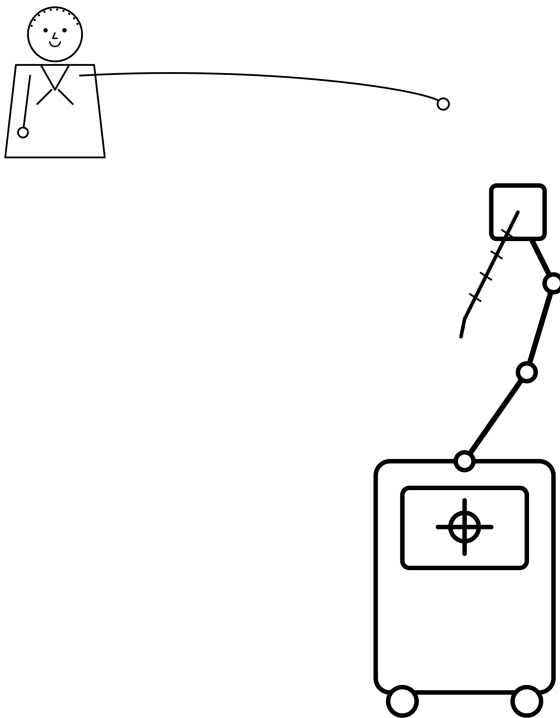
This robotic treatment couch precisely positions your body using six degrees of freedom to align with the radiation beam for your lung tumor treatment.

1. Lie on the robotic couch in the position practiced during simulation, place your arms in the overhead cradle, and allow the immobilization mask to be fitted over 2–3 minutes.
2. The couch shifts in 1–3 mm increments to align you with the beam; breathe normally and remain completely still during the 15–30 minute session, keeping your hands above your head.
3. When the beam stops, the couch lowers to exit height; wait for the immobilization device to be removed, then sit up slowly using both hands for support.

Patient-Robot Instructions: AI Oncology Trials - Needle-Placement Robots

Patient-Robot Interaction: Robotic Needle-Placement Systems

Adult Oncology Trial Setting



Robotic Needle-Placement Systems



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Robotic Needle-Placement Systems

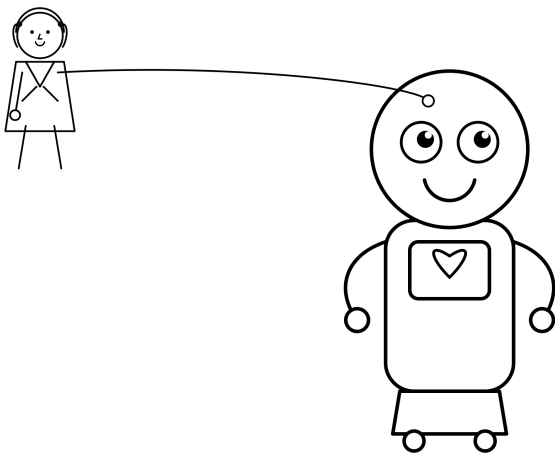
This robotic needle-guidance system uses real-time CT imaging to place a biopsy needle with sub-millimeter accuracy into your liver lesion.

1. Lie on the imaging table face-up with both hands at your sides, hold still while local anesthetic numbs the insertion site for 1–2 minutes, and keep your breathing shallow.
2. The robot aligns and inserts the needle over 1–3 minutes; hold your breath for 5–10 seconds when prompted for image capture during the 20–45 minute total procedure.
3. After the needle is withdrawn, press gently on the gauze placed over the site for 3–5 minutes, then remain lying down for 15–30 minutes of monitoring.

Patient-Robot Instructions: AI Oncology Trials - Companion Robots

Patient-Robot Interaction: Social Companion Robots

Pediatric Oncology Trial Setting



Social Companion Robots



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Social Companion Robots

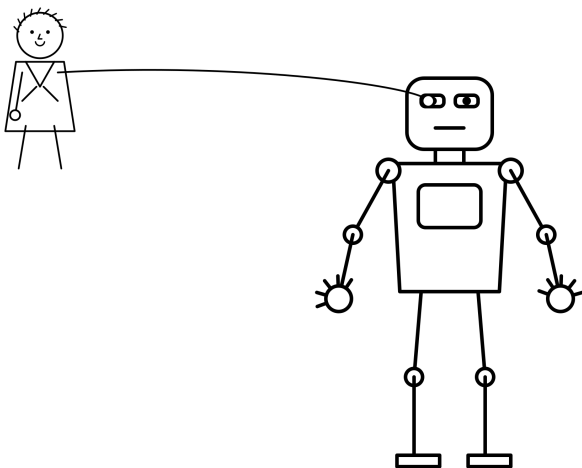
This friendly companion robot will talk and play interactive games with you to help you feel comfortable before your leukemia treatment today.

1. Walk into the room and wave or say “Hello” to the 1.2-meter tall robot; it will greet you by name and offer a gentle high-five with its soft hand if you feel ready.
2. Sit or stand near the robot and follow the breathing games on its chest screen for 10–20 minutes; talk to it normally and touch its hands or arms gently when it asks.
3. The robot says goodbye and shows a virtual sticker on its screen; give it a high-five or wave, then walk to the door where a staff member is waiting.

Patient-Robot Instructions: AI Oncology Trials - Humanoids

Patient-Robot Interaction: Humanoids

Pediatric Oncology Trial Setting



Humanoids



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Humanoids

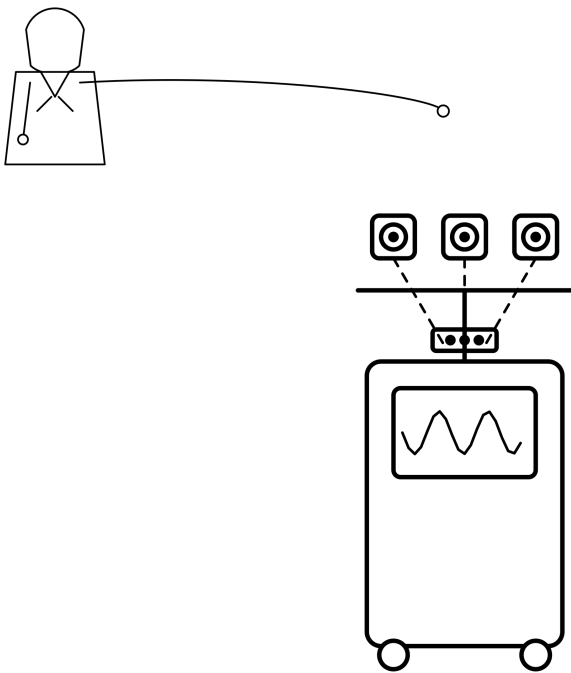
This humanoid robot will walk with you and practice the route to your treatment room to help you feel prepared for your bone cancer therapy session.

1. Enter the activity room and approach the 1.5-meter tall robot when it waves; hold its hand if comfortable—its grip is very gentle—and let it kneel to your eye level over about 1 minute.
2. Walk alongside the robot at your own pace (up to 1.0 m/s) for 3–5 minutes practicing the route, then follow its deep-breathing demonstration: inhale 1-2-3, exhale 1-2-3, for 15–25 minutes total.
3. The robot walks you back to the waiting area, waves, and says “Great job!”; release its hand and walk to the door where your parent or guardian is waiting.

Patient-Robot Interaction: AI Oncology Trials - RT Motion-Tracking Robots

Patient-Robot Interaction: Radiotherapy Motion-Management / Tracking Robots

Adult Oncology Trial Setting



Radiotherapy Motion-Management / Tracking Robots



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Radiotherapy Motion-Management / Tracking Robots

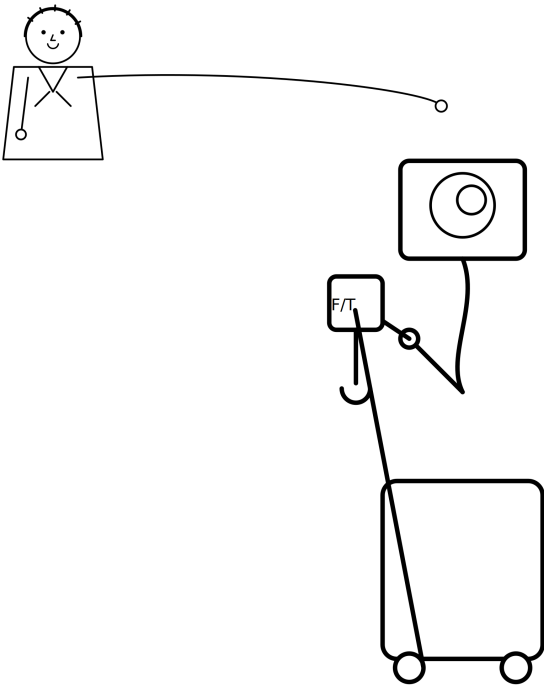
This motion-tracking system monitors your breathing in real time so the radiation beam activates only when your pancreatic tumor is precisely aligned.

1. Lie on the treatment couch with arms at your sides, allow a small reflective marker block to be placed on your abdomen, and breathe normally while the system calibrates in 1–2 minutes.
2. Follow the ceiling-screen breathing guide—inhale when the bar rises, exhale when it falls—for 10–20 minutes; remain still between breaths within the 2–3 mm tracking tolerance.
3. After the final beam, the system deactivates; hold still while the marker block is removed from your abdomen, then sit up slowly using both hands on the couch edge.

Patient-Robot Instructions: AI Oncology Trials - Imaging Robots

Patient-Robot Interaction: Imaging Assistant Robots

Adult Oncology Trial Setting



Imaging Assistant Robots



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Imaging Assistant Robots

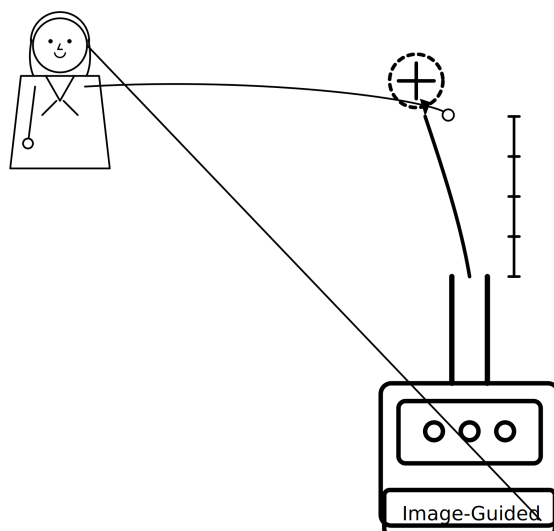
This robotic ultrasound system scans your neck area with consistent gentle pressure to evaluate your thyroid nodule and guide treatment planning.

1. Lie on the exam bed and tilt your head slightly back; place both hands at your sides and remain relaxed while ultrasound gel is applied to your neck over approximately 1 minute.
2. The robotic probe applies 1–3 N of gentle, steady pressure and scans in a grid pattern for 10–20 minutes; breathe normally as the robot compensates for motion automatically.
3. The probe lifts away and excess gel is wiped with a warm cloth; you may sit up immediately—no recovery period is needed.

Patient-Robot Instructions: AI Oncology Trials - Steerable Needle Robots

Patient-Robot Interaction: Steerable Needle / Needle-Steering Robots

Adult Oncology Trial Setting



Steerable Needle / Needle-Steering Robots



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Steerable Needle / Needle-Steering Robots

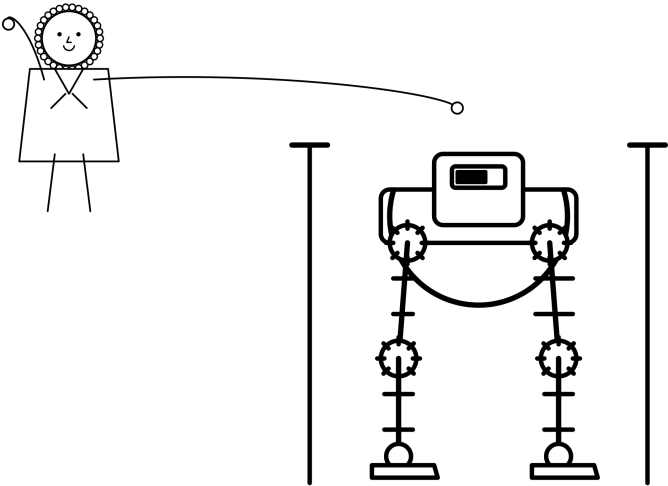
This needle-steering robot curves a flexible needle along a pre-planned path through tissue to reach your kidney tumor for targeted ablation treatment.

1. Lie on the interventional table with pillows for support, place both hands above your head, and hold still while local anesthetic is injected at the entry site (brief 5-second sting).
2. The steerable needle advances along a curved path while real-time MRI confirms position every 5–10 seconds; remain motionless for the 30–60 minute procedure and breathe shallowly.
3. The needle is slowly withdrawn; apply gentle pressure to the gauze on the entry site for 5–10 minutes, then rest on the table for 30–60 minutes of monitoring.

Patient-Robot Instructions: AI Oncology Trials - Rehab Exoskeletons

Patient-Robot Interaction: Rehabilitation Exoskeletons / Robotic Gait Trainers

Adult Oncology Trial Setting



Rehabilitation Exoskeletons / Robotic Gait Trainers



Page 10 of 10

Rehabilitation Exoskeletons / Robotic Gait Trainers

This lower-body exoskeleton supports your legs and guides your walking motion to rebuild strength and mobility after your bone cancer surgery.

1. Stand with your back to the exoskeleton frame, grip both parallel bars firmly, and hold steady while waist, thigh, shank, and foot straps are secured over 2–3 minutes.
2. Follow the audible step cues and walk between the parallel bars at 0.2–0.5 m/s for 15–30 minutes; say “Pause” to stop at any time, keeping both hands on the bars throughout.
3. The exoskeleton returns you to a standing rest position; keep gripping the bars while straps are removed in reverse order (feet, shank, thigh, waist), then sit and rest for 5 minutes.